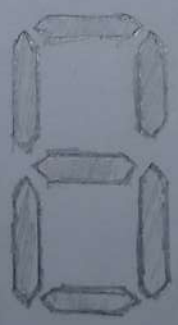
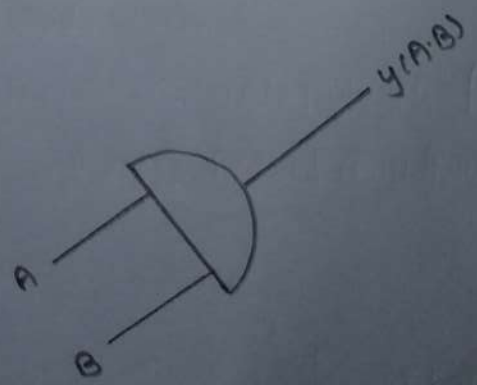
Study of Digital Electronic Component's

Components:

- Registor
- Transistor
- Switch
- LED
- Power Supply
- Digital IC
- Logic Gates
- 7 segment decoder
- Banana Cable
- frequency generator
- Digital Trainer kit
- Bread Bard
- Vcc
- Ground.
- Digital clock





THEORY

- Digital electronics deals with circuit that use two discrete levels. HIGH(1) and LOW (0). It forms the basis of modern electronic devices and computers. Components like resistors, transistors, LEDs, ICs, Logic gates etc are used to build digital circuits. Logic gates perform logical operations, while ICs combine multiple components in a single chip. Using a breadbone or digital trainer, we can easily connect components and observe the working of logic circuits.

Description of Components

Resistor

A resistor is a passive component that helps control the flow of electricity in a circuit. It keep things, like LEDs from getting damaged due to excess current. The value of resistance is measured in ohms (Ω). Resistors are commonly used in analog and digital circuit.

fig: Resistor

Transistor

A tansistor is a semiconductor device used as a switch or an amplifier. In digital electronics, it mainly works as a switch. It has three terminals: **emitter**, **base** and **collector**. Transistor are widely used in logic circuits and ICs.

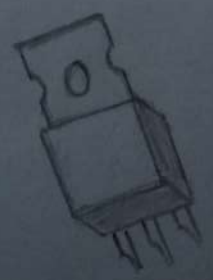
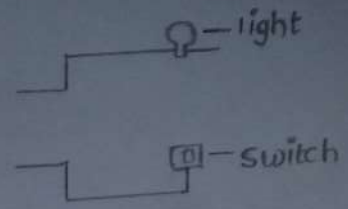


fig: Transistor

Switch



A switch is any device that is used to interupt the flow of electrons in a circuit and also connects other devices together. A switch is used to open or close an electrical circuit. It controls the flow of current. It controls the flow of current. In digital circuits, switches are used to provide ~~signal~~ input signals. switches help in testing different input combinations.

Light Emitting Diode (LED)

An LED is a semiconductor device that emits light when forward biased. It is commonly used as an indicator in digital circuits. LED grows when correct voltage and polarity are applied. A resistor is always used with an LED to limit current. LEDs were used as indicators to observe the output of logic gates on the breadboard.

fig: LED

Power supply

A power supply provides the required voltage and current to the circuit. It converts electrical energy into usable form. Proper voltage is necessary for correct operation of digital components. Incorrect voltage may damage components.

fig: Power supply

Digital IC

An integrated circuit is a small electronic chip containing components like resistors, transistors, and logic gates. ICs reduce circuit size and power consumption. They improve reliability and efficiency. logic gate ICs are commonly used in digital electronics.

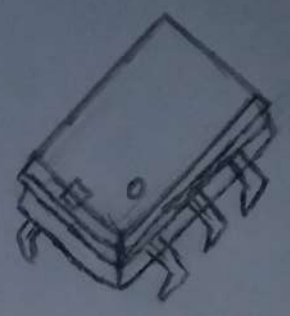


fig: Digital IC

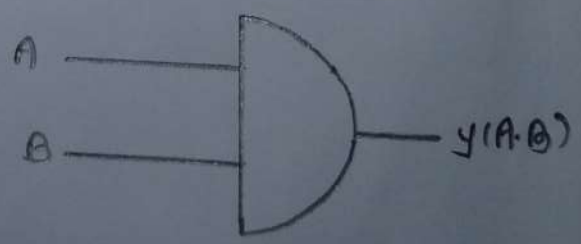
Logic Gates

logic gates are the fundamental building blocks of digital circuits, acting as tiny electronic switches that perform basic logical operations (like AND, OR, NOT) on one or more binary inputs (0s and 1s) to produce a single binary output.

Different types of gates:

AND Gate

An AND gate give HIGH output only when all inputs are HIGH. If any input in LOW, the output becomes LOW.

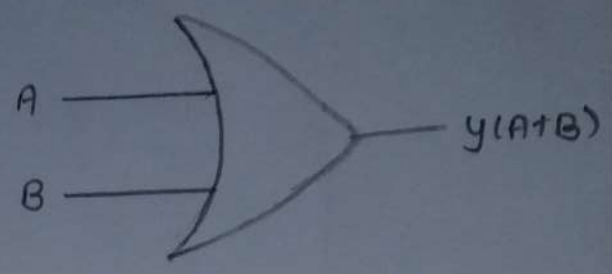


Truth table

Input		Output
A	B	A·B
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate

An OR gate give HIGH output when any one of the inputs is HIGH. The output is LOW only when all inputs are LOW.

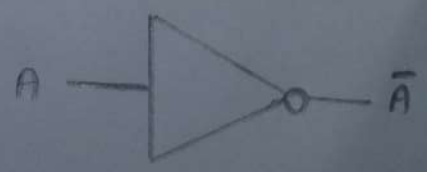


Truth table

Input		Output
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate

A NOT gate gives the opposite output of the input. If input is HIGH, output is LOW and vice-versa.

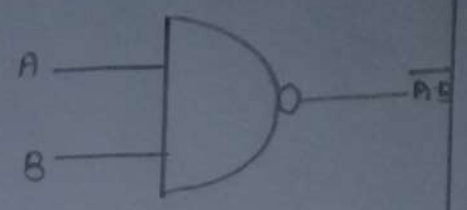


Truth table

Input	Output
A	A'
0	1
0	1
1	0
1	0

NAND Gate

A NAND Gate is the inverse of AND Gate. It gives LOW output only when all inputs are high.

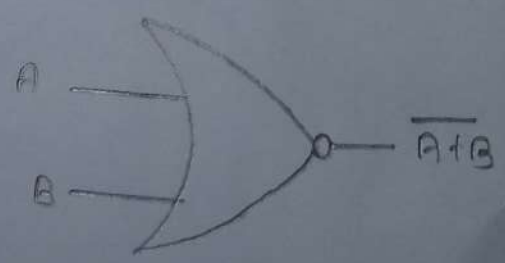


Truth table.

Input		Output
A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

A NOR gate is the inverse of OR gate. It gives HIGH output only when all inputs are low.

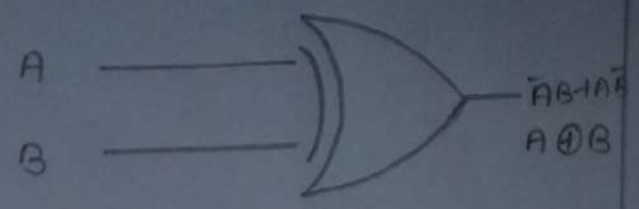


Truth table

Input		Output
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate

An XOR Gate gives HIGH output when inputs are different. If both inputs are same, output is low.

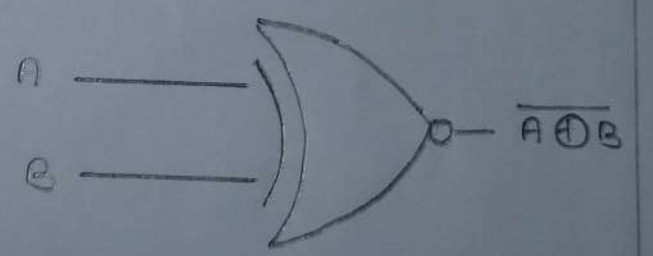


Truth Table

Input		Output
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

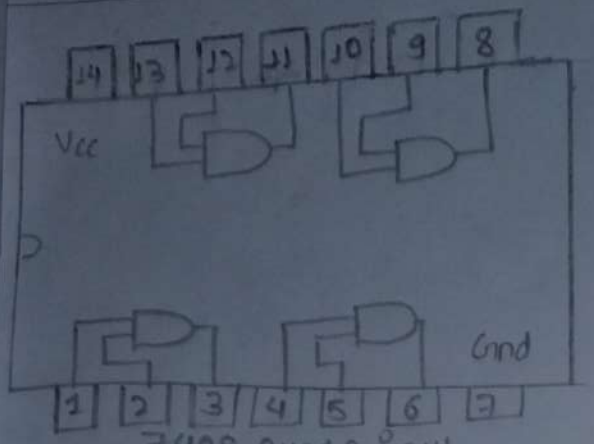
XNOR Gate

An XNOR Gate gives HIGH output when inputs are the same. It gives low output when inputs are different.

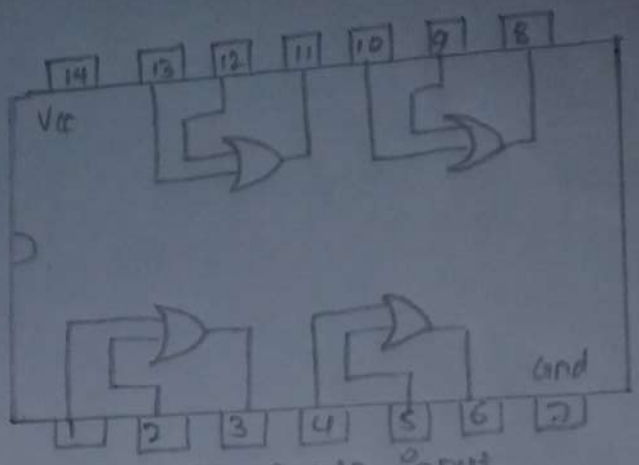


Truth Table

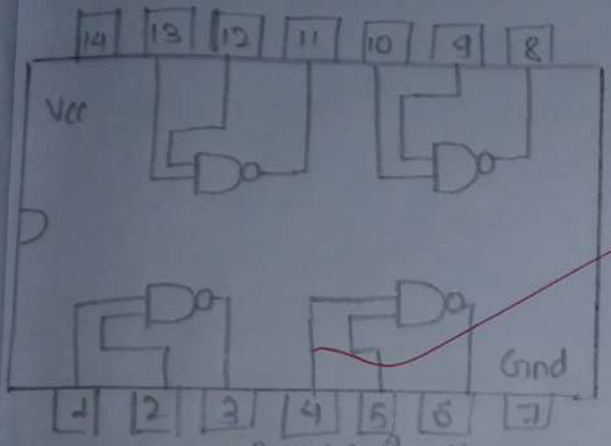
Input		Output
A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1



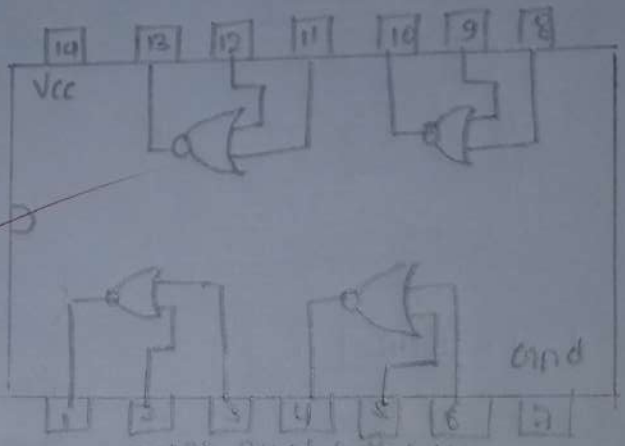
7408 Quad 2 input AND Gates



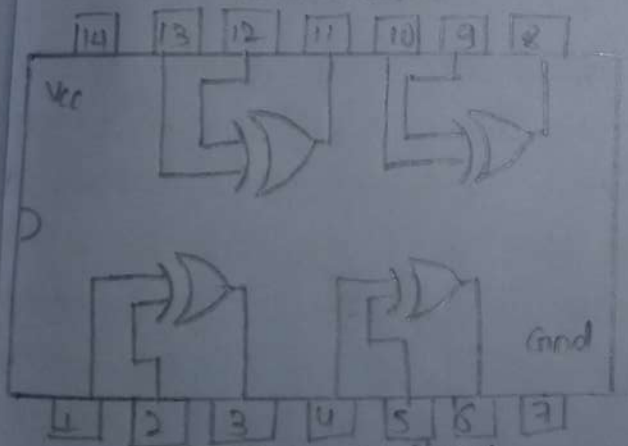
7432 Quad 2 input OR Gates



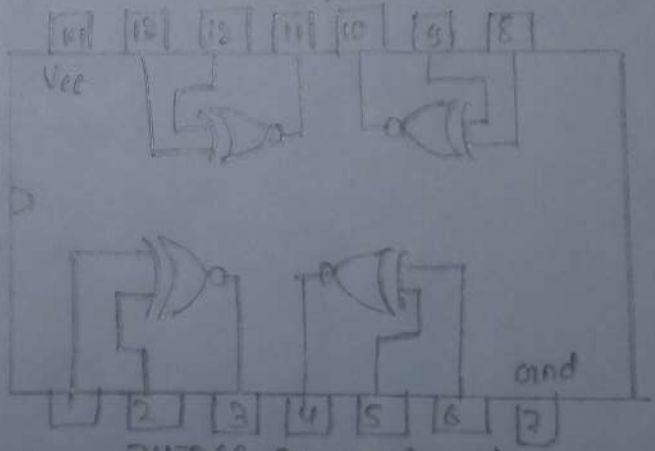
7400 Quad 2 input NAND Gates



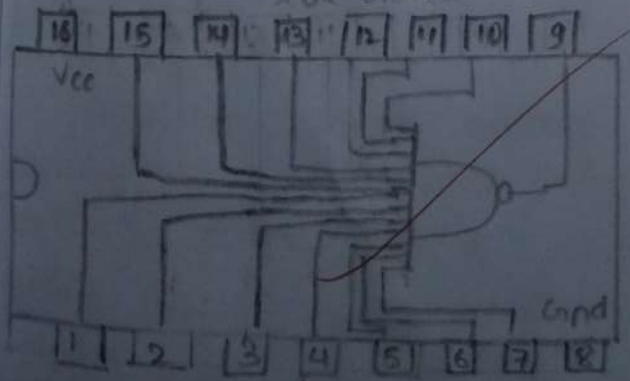
7402 Quad 2 input NOR Gates



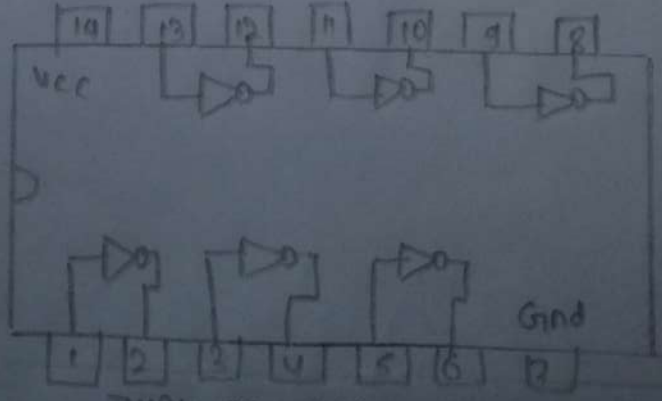
7486 Quad 2 input XOR Gates



74266 Quad 2 input XNOR Gates



74133 single 13 input NAND Gates



7404 Hex NOT Gates (Inverters)

7 segment Decoder

A 7-segment decoder is a digital circuit of IC that converts binary inputs into signal to control a 7 segment display. It determines which segments should glow to display the correct decimal number. By giving a binary number as input, the decoder lights up the corresponding segment on the display. It makes it easier to display numbers without manually connecting each segment.

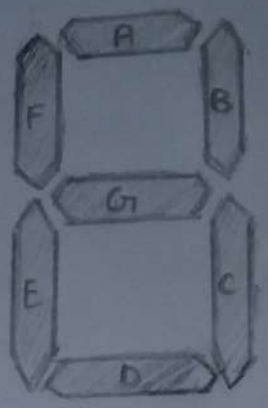


fig : 7-segment decoder

Banana Cable

Banana cables are used to make electrical connections between different digital electronic components in a circuit. They allow the flow of current and digital signals from one component to another. In digital electronics, these cables are commonly used on bread boards and digital trainer. Proper wiring ensures correct operation of the circuit.

fig : Banana Cable

Frequency generator.

A frequency generator creates precise, controllable digital pulse (like square wave) for testing logic circuits, often using Direct Digital synthesis (DDS) or simple RC oscillators with schmitt triggers to generate adjustable frequencies.

fig : clock

Digital Trainer kit

A digital trainer kit is a learning kit used to perform experiments in digital electronics. It has a built-in power supply, switches, LEDs and IC sockets for connecting components. Using switches as input and LEDs as outputs, the working of logic gates and other circuits can be easily observed. Digital trainer kit make learning digital electronics simple and safe.

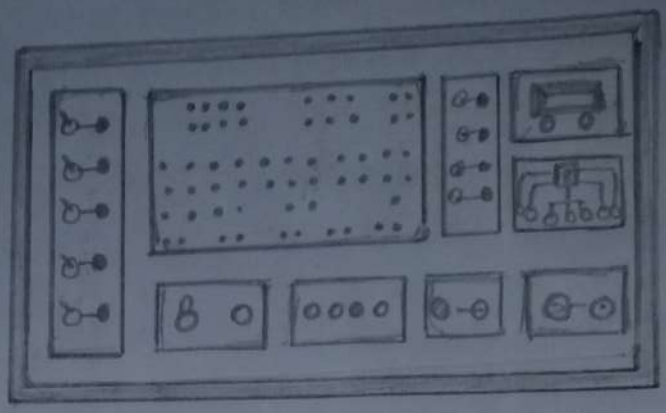


fig: Digital Trainer kit.

Bread Board

A bread board is used to make temporary electronic circuit without soldering. It allows easy connection and removal of components. Breadboards are useful for testing and learning purposes. They help in quick circuit assembly.

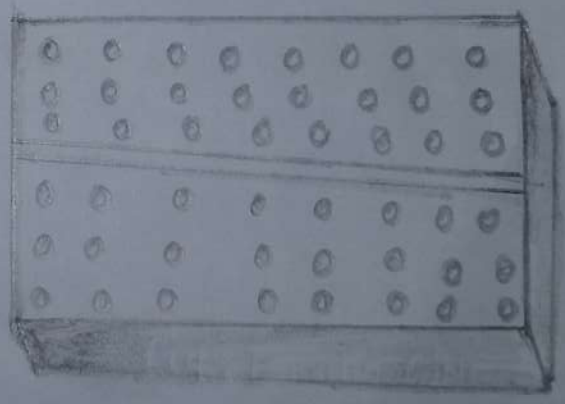


fig: Bread Board

Vcc

Vcc (voltage at common collector) is the standard notation for the positive power supply voltage that energizes integrated circuits (ICs) and digital components, typically +5V for older Transistor-Transistor Logic (TTL) or lower like 3.3V for modern devices.

Ground

Ground is the essential 0-volt reference point for voltage measurements, acting as the common return path for ~~voltage~~ current, providing signal stability, and defining logic levels (like '0') for consistent circuit operation, usually connected to the negative terminal of the power supply.

Digital clock

A digital clock is an electronic device that displays time in numerical form. It works using clock pulses generated by an internal oscillator. These pulses control counting circuits to measure seconds, minute and hours. Digital clock are commonly used in electronic systems for accurate time measurement. They are widely used in watches, timers and digital clock.

fig: Digital clock

Conclusion

Thus, we successfully studied and became familiar with various digital electronic components. We also learned the practical working of logic gates by connecting them on a digital trainer kit and observing the output through LED. This experiment improved our understandings of digital electronics and basic logic operations.